

Tan Xin Er

Software Engineer

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PROFESSIONAL SUMMARY

Software Engineering graduate with hands-on experience developing web, mobile, and IoT applications. Experienced in building AI-powered applications, automation tools, and intelligent systems. Passionate about full-stack development, scalable systems, and leveraging artificial intelligence to solve real-world problems.

EDUCATION

Bachelor of Engineering in Information & Communications Technology
majoring in Software Engineering (Honours with Merit) | May 2026

Singapore Institute of Technology (SIT), Singapore

Diploma in Accountancy | April 2022
Singapore Polytechnic (SP), Singapore

WORK EXPERIENCE

IT Application Intern

SBS Transit Pte Ltd

Singapore

May 2025 – April 2026

- Developed SmartTDG (https://tanxiner.github.io/other_projects/smarttdg.html), an offline AI-powered technical documentation generator for legacy .NET applications using Python, FastAPI, React, Roslyn Analyzer, LangChain, Ollama, and local LLMs, enabling automated code understanding and documentation generation.
- Containerized the full application stack using Docker to support offline and secure deployment in restricted environments, packaging backend services, AI models, and required dependencies for consistent execution.
- Built a multi-language static analysis pipeline using C#, VB.NET, JavaScript, HTML, SQL parsers, and PlantUML/Mermaid to extract code structures, dependencies, relationships, and database interactions.
- Automated generation of technical documentation, architecture diagrams, and functional summaries through AI-assisted analysis, reducing manual documentation effort for complex legacy systems.
- Developed backend services for internal enterprise systems using .NET (C#, ASP.NET) and SQL, implementing business logic, stored procedures, and integrations to streamline workflows, support feature enhancements, and ensure database reliability.

Data Engineer Intern

Munich Re

Singapore

December 2024 – April 2025

- Developed and optimized R-based ETL pipelines to automate data processing workflows, reducing processing time by over 50% while improving data integration, reporting efficiency, and maintainability of existing scripts.

Junior Research Assistant Intern

Illinois Advanced Research Centre

Singapore

May 2024 – July 2024

- Designed and implemented a web-based security analytics dashboard for cyber-physical systems, enabling centralized monitoring and visualization of IoT and ICS environments including robotics, microgrids, and building management systems.
- Developed data integration pipelines and visualization modules to aggregate distributed industrial system data, map system relationships, monitor security indicators, and provide actionable cybersecurity insights for improved situational awareness.

PROJECTS / EXPERIENCES

Semantic Segmentation Mobile Application in Autonomous Driving

(https://tanxiner.github.io/other_projects/roadvisionai.html)

- Developed a real-time semantic segmentation pipeline using TensorFlow, training and optimizing BiSeNet and MobileNet-UNet architectures specifically for mobile edge deployment.
- Built a custom, localized dataset of Singaporean urban landscapes using CVAT, applying data augmentation techniques to improve model generalization and regional accuracy over standard datasets like CityScapes.

Real-Time Multi-Modal Assistive Intelligence System (https://tanxiner.github.io/other_projects/assistededge.html)

- Developed a distributed Raspberry Pi edge system for real-time ASL fingerspelling, speech-to-text (STT), and facial emotion recognition, integrating computer vision and speech recognition models with a React-based dashboard for real-time interaction.
- Engineered low-latency MQTT and Socket.IO architecture streaming sensor and AI data from edge devices to backends for real-time processing and visualization.

Secure Backend System with AWS IAM Authentication

- Executed and tested a RESTful API to support backend data retrieval and business workflows, implementing SQL based data handling and validation logic.
- Secured API endpoints using AWS IAM (users, roles, and policies) and built a C# console application to consume and validate authenticated endpoints end-to-end.

GitLab CI/CD Pipeline Automation

- Designed and implemented a GitLab CI/CD pipeline to automate software build, testing, and deployment workflows across development environments and production releases, improving development efficiency, deployment consistency, and overall software delivery reliability.

SKILLS

Python, R, JavaScript, TypeScript, SQL, Flask, React, REST APIs, NLP, LLMs, Gen AI, LangChain, AWS, Git, Docker

AWARDS / HACKATHONS

SBS Hackathon 2025 - 1st Runner Up

- Created an ESP8266 LED control system using MQTT for real-time remote brightness adjustment, optimizing power consumption and display visibility.

CERTIFICATIONS

- SCRUM Master
- Microsoft Office Excel & Excel Expert

LANGUAGES

English: Fluent

Chinese: Intermediate

Bahasa Indonesia: Basic